

Lesson 3: Is a force real?

The Brake · Newton's first and second laws · 3 sessions of 50 minutes

Grade 11 scientific track (adaptable to Grade 10 and to equivalent grades in other Arab countries). Physics in English or French; discussion may be in Arabic. The student lesson is free at sababphysics.com: Lessons, Grade 11, Lesson 3.

Learning objectives

- State and apply Newton's first and second laws.
- Derive $a = \text{net force} / m$ from experimental data.
- Distinguish applied force, friction and net force.
- Argue a position on what a force is, using evidence.

Competencies of the 2026 framework

Competency	Where it is developed
Critical thinking	Steps 2 and 6: position, argument, position shift
Inquiry	Steps 3 and 4: collect data, find the law
Digital	Virtual lab and data table
Communication and collaboration	Community of inquiry, written reflection
Creativity and innovation	Step 5: Arduino lab or egg-passenger challenge

The six steps at a glance

1 Hook	Film "The Brake" and the taxi simulation. Students predict what pushes the passenger forward.
2 Question	"Is a force real?" Four thinkers, four answers. Students take a position.
3 Investigate	Virtual cart: vary the hanging mass and the cart mass, collect accelerations.
4 Model	Test four models against the data; Newton's second law emerges. Back to the taxi.
5 Make	Arduino lab (measure a real acceleration) or the egg-passenger challenge.
6 Reflect	Position shift, concept ladder, written reflection.

Session timeline

S1 · 0'	Hook: play the film and the taxi animation on the projector. Students predict individually, then vote.
S1 · 10'	Community of inquiry (circle): "Is a force real?" Read the four thinkers. Roles: facilitator, questioner, summariser, sceptic.
S1 · 35'	Students write their position and one reason (step 2). Homework: 3 lab runs on a phone.
S2 · 0'	Groups compare data and test the four models (step 4). Extension: friction runs and net force.
S2 · 25'	Explain the taxi with the first law. Exam-style practice.
S3 · 0'	Make (step 5): half the class runs the Arduino lab, half builds the egg-passenger cart. Groups swap findings at the end.
S3 · 40'	Reflection and position shift (step 6). Students send their reflection to the class group.

The four thinkers (step 2)

Aristotle

4th century BCE

A moving body needs a mover. Without a continuing push, motion stops.

Ibn Sina (Avicenna)

11th century

A thrown body receives an inclination (mayl) from the thrower that keeps it moving until resistance wears it away.

Isaac Newton

1687

A body keeps its state of rest or of straight, uniform motion unless a net force acts.
Force changes motion: $\text{net force} = m \cdot a$.

Ernst Mach

19th century

Force is not a hidden thing: it is a name for how bodies change each other's motion. A description.

Short summaries for discussion, not quotations.

Facilitation moves

- "What do you mean by...?" (clarify)
- "How do you know?" (evidence)
- "Can anyone give a counter-example?"
- "What would Ibn Sina say to Aristotle?"
- Do not give the answer before step 4.

Common misconceptions and how to challenge them

Motion needs a force.

Ask: what keeps a puck sliding on ice? Show the virtual cart with no friction.

A moving object "has" a force that runs out.

Compare with Ibn Sina's mayl, then ask what the data from step 3 shows.

Heavier objects always accelerate less, whatever the force.

Point to runs where the force changes with the mass.

The forward "push" felt when braking is a real force.

Ask: what object exerts it? No object, no force: the passenger keeps moving while the taxi slows.

Make step: Arduino lab

- Per group: Arduino Uno, HC-SR04 distance sensor, 4 jumper wires, toy car, string, small water bottle.
- Students paste the Serial Monitor data; the platform fits $x(t)$ and compares a with the theory.
- No kits? Use the Wokwi online simulator or the built-in sample data.
- Discussion: when theory and measurement disagree, which one is "wrong"?

Make step: egg-passenger challenge

- Recycled materials only: bottles, cardboard, sponges, tape.
- Teams predict with the on-screen calculator, build, test, then improve once.
- Score: egg intact (3), prediction matched the test (2), design journal uses $F = m \cdot a$ (3), poster (2).
- Citizenship link: display the posters in real service taxis.
- Safety: raw eggs, low ramps, clean up spills at once.

Assessment rubric: written argument (steps 2 and 6)

Criterion	1	2	3
Claim	Unclear	Clear	Clear and nuanced
Evidence	None	Daily-life example	Uses own lab data
Reasoning	Restates	Links evidence to claim	Uses Newton's laws correctly
Counter-argument	Absent	Mentioned	Answered

Student worksheet · Lesson 3

Photocopy for each student

Name: _____ Class: _____ Date: _____

1. Predict: what pushes the passenger forward when the taxi brakes?

- ☐ A The braking pushes the passenger forward.
- ☐ B Nothing pushes them. They keep moving while the car slows down.
- ☐ C Gravity pulls them forward.
- ☐ D The air inside the car pushes them.

2. My data (step 3)

Run	Hanging mass (g)	Cart mass (g)	Acceleration (m/s^2)	Net force (N)
1				
2				
3				
4				
5				

3. My position: is a force a real thing, or only a description?

A real thing

Only a description



Mark B (before, step 2) and A (after, step 6) on the line.

4. Exit ticket

A passenger lurches forward when a service taxi brakes suddenly. Explain why, using only real forces (forces exerted by an object).
